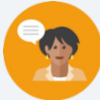


## 7.7 Finding the Area of Polygons

### Lesson Introduction

|                                                                                                       |                                                                                                                                                                                                                               |                                                                                                        |                                                                                                                           |                                                                                                                                                                                                                  |                                                                                                                                                                                                                      |
|-------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  Benchmark           | MA.7.GR.1.2 Solve mathematical or real-world problems involving the area of polygons or composite figures by decomposing them into triangles or quadrilaterals.                                                               |                                                                                                        |  Learning Targets                      | <ul style="list-style-type: none"><li>I can decompose a regular polygon into triangles to find the area.</li><li>I can decompose a regular polygon into triangles and quadrilaterals to find the area.</li></ul> |                                                                                                                                                                                                                      |
|  Benchmark Coherence | Building On<br>MA.6.GR.2.2 Solve mathematical and real-world problems involving the area of quadrilaterals and composite figures by decomposing them into triangles or rectangles.                                            |                                                                                                        | Working Toward<br>MA.912.GR.4.4 Solve mathematical and real-world problems involving the area of two-dimensional figures. |                                                                                                                                                                                                                  |                                                                                                                                                                                                                      |
|  Essential Questions | <ul style="list-style-type: none"><li>How do you decompose regular polygons into triangles and quadrilaterals to find the area?</li><li>How do you decompose regular polygons into triangles to find the area?</li></ul>      |                                                                                                        |                                                                                                                           |                                                                                                                                                                                                                  |                                                                                                                                                                                                                      |
|  Lesson Narrative    | Students derived the area formulas for trapezoids, parallelograms, and rhombi in previous lessons. In this lesson, students decompose regular polygons into only triangles or triangles and quadrilaterals to find the areas. |                                                                                                        |                                                                                                                           |                                                                                                                                                                                                                  |                                                                                                                                                                                                                      |
|  Component Summary | 7.7.1 Warm-Up (~5 min.)                                                                                                                                                                                                       | Justify which rectangle is more like a square ( <i>SE p. 332</i> )                                     | 7.7.3 Collaboration (10 min.)                                                                                             | Collaboratively decompose regular pentagons and hexagons to find areas ( <i>SE p. 335</i> )                                                                                                                      |                                                                                                                                                                                                                      |
|                                                                                                       | 7.7.2 Guided Instruction (15 min.)                                                                                                                                                                                            | Decompose regular hexagons to find areas ( <i>SE p. 333</i> )                                          | 7.7.4 Your Turn! (10 min.)                                                                                                | Decompose regular polygons to find areas ( <i>SE p. 336</i> )                                                                                                                                                    |                                                                                                                                                                                                                      |
|                                                                                                       | 7.7.5 Wrap-Up (~5 min.)                                                                                                                                                                                                       | Identify and correct errors in a sample of student work finding area of a nonagon ( <i>SE p. 337</i> ) |                                                                                                                           |                                                                                                                                                                                                                  |                                                                                                                                                                                                                      |
|  Resources         | Glossary                                                                                                                                                                                                                      |                                                                                                        | Materials                                                                                                                 |                                                                                                                                                                                                                  | Additional Preparation                                                                                                                                                                                               |
|                                                                                                       | <u>Key Terms:</u> <ul style="list-style-type: none"><li>Regular polygon</li></ul> <u>Additional Terms:</u> N/A                                                                                                                |                                                                                                        | <ul style="list-style-type: none"><li>(Optional) Pattern blocks</li><li>(Optional) Highlighters</li></ul>                 |                                                                                                                                                                                                                  | <ul style="list-style-type: none"><li>Consider placing the pattern blocks needed to model the decomposition of each polygon into a labeled bag or container for each student to use throughout the lesson.</li></ul> |

|                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                               |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |  |
|------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| <br>Teacher Tips                | Common Misconceptions                                                                                                                                                                                                                                                                                                                                                                                         |  | Things to Consider                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |  |
|                                                                                                                  | <ul style="list-style-type: none"><li>Students may incorrectly think that the area of regular polygons decomposed into smaller non-congruent triangles can be found by multiplying the area of one triangle by the number of triangles.</li><li>Students may not include the units with their answers or may not raise the units to the power of 2.</li><li>Students may round mid-step.</li></ul>            |  | <ul style="list-style-type: none"><li><i>Students using the Workbook may benefit from having multiple copies of each shape to spend less time erasing or drawing, and more time making mistakes and refining their approaches to decomposition.</i></li><li>Consider allowing students to use a calculator for the lesson. In problems that require several steps students should not round until the final step. Rounding mid-step could significantly change the answer. If a place value is not given for rounding, then determine what place value you would like your students to round to. Many assessments have students round to three decimal values. Incorporating the Desmos scientific calculator can add a powerful tool for students who struggle with completing steps on a handheld calculator. The Desmos calculator allows students to input complicated arithmetic that would require several steps on a handheld calculator. This eliminates common careless errors that many students make.</li><li>If time is an issue, then consider assigning 7.7.4 Your Turn! for homework.</li></ul> |  |
|                                                                                                                  | Literacy and Language Routines                                                                                                                                                                                                                                                                                                                                                                                |  | Mathematical Thinking and Reasoning (MTR) Standard                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |  |
|                                                                                                                  | <b>Stronger and Clearer</b><br>7.7.1 Warm-Up asks students to identify which given rectangle is more like a square. Ask students to write a justification for their selection and share it with a partner. Ask the partner to provide feedback. Next, give students time to revise their justifications based on the feedback that they received. Finally, repeat the sharing, feedback, and revising cycle.  |  | <b>MA.K12.MTR.2.1: Multiple Representations</b><br>Students decompose regular polygons into only triangles or triangles and parallelograms to determine their areas. Students work with concrete representations of the area formula for the polygons. Through the activities, students are guided from concrete, to pictorial, to abstract representations as their understanding of the areas progresses.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |  |
| <br>Differentiate Instruction | Consider having On-Ramp to 6 <sup>th</sup> Grade videos available for students to use as additional scaffolding during the lesson or to use as a “preview” before the lesson to practice their fluency with finding area. <ul style="list-style-type: none"><li>Finding Area of Figures Using Unit Squares</li><li>Finding Area of Rectilinear Shapes</li><li>Using the Area Formula for Rectangles</li></ul> |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |  |
| Lesson Notes:                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                               |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |  |

## 7.7.1 Warm-Up: Which One?

**Component Overview:** Students are provided with the dimensions for two different rectangles and asked to explain which is more like a square. The instructional routine “Stronger and Clearer Each Time” provides an auditory entry point to help students refine their justifications.



### 7.7.1 Warm-Up: Which One?

Two rectangles are given.

#### Rectangle A

13 units by 15 units

#### Rectangle B

2 units by 3 units

1. Explain which rectangle is more like a square.

#### Rectangle A

Answers will vary. Since all of the sides of a square are the same length, the side length divided by the side length will equal 1. Dividing the side lengths of Rectangle A will give an answer closer to 1 than dividing the sides of Rectangle B.



#### Stronger and Clearer Each Time

Ask students to write a justification for their selection and share it with a partner. Ask the partner to provide feedback. Next, give students time to revise their justification based on the feedback they received. Finally, repeat the sharing, feedback, and revising cycle.

## 7.7.2 Guided Instruction: Decomposing Regular Polygons

**Component Overview:** Students explore how to decompose a regular octagon into either congruent triangles, or triangles and quadrilaterals. Students compare the areas and discover that they are the same for both decomposition methods.

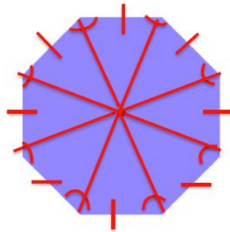


### 7.7.2 Guided Instruction: Decomposing Regular Polygons



A **regular polygon** is a polygon that is both equilateral (all sides congruent) and equiangular (all angles congruent).

1. A regular octagon is shown.



- a. On the figure, mark all congruent angles and all congruent sides.  
*See diagram above.*
- b. Draw lines to decompose the regular octagon into congruent triangles.  
*See diagram above.*
- c. How many congruent triangles did you decompose the regular octagon into?  
**8**
- d. Discuss with your partner how decomposing the octagon into congruent triangles is helpful in determining the area.  
*Answers will vary. The area of the octagon will be the same as the sum of the area of all of the decomposed triangles.*



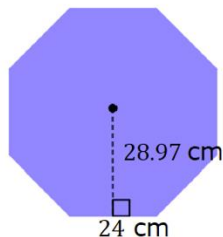
Students could decompose the octagon into 16 or 8 congruent triangles. Engage students in a discussion about which decomposition is correct and why. Prompt students to recognize that because all the triangles are congruent, they can multiply the area of one of the triangles by the number of triangles.



Consider providing students with pattern blocks to decompose the shapes throughout the lesson. This strategy could be replicated with paper and scissors.

## 7.7.2 Guided Instruction: Decomposing Regular Polygons (continued)

2. Some dimensions of an octagon are given in centimeters (cm). Each side of the octagon has a length of 24 cm. The perpendicular distance from the center to midpoint of a side is 28.97 cm.



- a. Find the area of one decomposed triangle. Round your answer to the nearest thousandth.

$347.64 \text{ cm}^2$

- b. Find the area of the regular octagon rounded to the nearest thousandth of a square centimeter ( $\text{cm}^2$ ) by finding the sum of the areas of all the decomposed triangles.

$2,781.12 \text{ cm}^2$



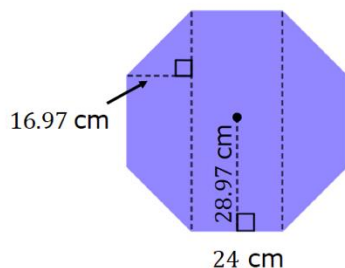
- Is this a regular octagon?
- How do you know?
- Would it be possible to find the area if this was not a regular octagon? Why or why not?



The answer to question 2a will depend on if students create 8 isosceles triangles or 16 right triangles. If a student decomposed the octagon into 16 right triangles, then the area of one decomposed triangle is  $173.82 \text{ cm}^2$ . Notice all questions in this lesson ask for students to round to the nearest thousandth for consistency. However, the answer here only requires rounding to the hundredths place.

## 7.7.2 Guided Instruction: Decomposing Regular Polygons (continued)

3. An octagon is divided into two trapezoids and a rectangle. Each side of the octagon has a length of 24 cm. The perpendicular distance from the center to midpoint of a side is 28.97 cm. The height of each trapezoid is 16.97 cm.



- a. Find the area of each decomposed quadrilateral. Round to the nearest hundredth.
- b. Using the decomposed shapes, determine the area of the octagon rounded to the nearest thousandth of a square centimeter ( $\text{cm}^2$ ).

$695.26\text{cm}^2$ ;  $1390.56\text{cm}^2$

$2,781.082\text{cm}^2$

4. Complete the chart using the areas from 2b and 3b

| Decomposition  | Area of Octagon       |
|----------------|-----------------------|
| triangles      | $2,781.12\text{cm}^2$ |
| quadrilaterals | $2,781.08\text{cm}^2$ |

5. What do you notice about the areas?
- Answers will vary. I notice that the areas are almost the same.*



Consider providing students with pattern blocks to decompose the shapes throughout the lesson. This strategy could be replicated with paper and scissors.



Consider an inquiry-based approach when exploring question 3 with students.

- How did you find the lengths of the dimensions that were not given in the original drawing?
- Do you need to calculate the area of the trapezoid on the left and on the right?
- Would the areas be different when the octagon is decomposed into triangles versus when it is decomposed into triangles and quadrilaterals?



Help students understand how to put all the work for 3b into the Desmos calculator as

$$16.97(2 \cdot 28.97 + 24) + (2 \cdot 28.9) \cdot 24.$$

Explain why the  $\frac{1}{2}$  is not necessary.

Students may wonder why the areas are slightly different. This occurs because the height of the trapezoid, 16.97, is a rounded value. This is a good opportunity to discuss which of these would be considered more accurate and how rounding can affect the outcome. Although this rounding was not something that the student did, it may be a good lesson on the dangers of rounding calculations along the way.



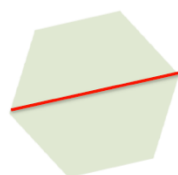
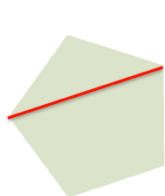
## 7.7.3 Collaboration: Decomposing Regular Polygons to Find Area

**Component Overview:** Students work with partners to decompose a regular pentagon and a regular hexagon into triangles and quadrilaterals to determine their areas. Question 1 should be completed individually by each student. Then, students work with partners for the remainder of the activity.



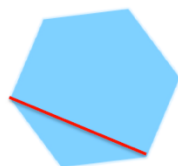
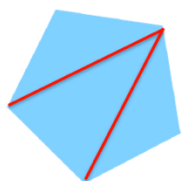
### 7.7.3 Collaboration: Decomposing Regular Polygons to Find Area

1. Decompose each of the following regular polygons.



*Answers will vary.*

2. Share with your partner how you decomposed your shapes. If your partner decomposed the shapes differently, draw on the blue shapes how they decomposed their shapes. If you decomposed the shapes in the same way, work together to create a different way to decompose the shapes.



Consider using question stems for students to guide their thinking while they collaborate or using these stems for pulling a small group during this activity.

Are there decompositions that would make it easier to find the area of the regular polygons?

If you decomposed the pentagon into triangles, are the triangles congruent?

If the triangles are not congruent, can you still use them to calculate area?

If you decomposed the hexagon into two trapezoids, are these trapezoids congruent? If you decomposed the hexagon into triangles, how many did you draw? Are they congruent?

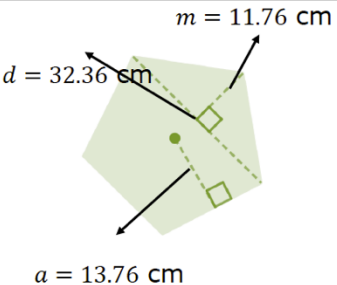
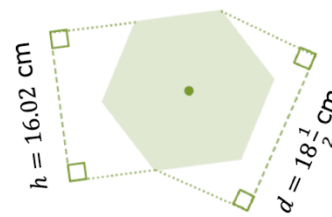
Choose one decomposition that you think would be easiest to use to find area. What lengths do you need to determine to find the areas?



If students decompose the pentagon into three triangles, they may incorrectly assume that the triangles are congruent. Prompt students to recognize that the three triangles are not congruent.

## 7.7.3 Collaboration: Decomposing Regular Polygons to Find Area (continued)

3. The table shows a pentagon and a hexagon. Given measurements in centimeters (cm). Work with your partner to determine the area rounded to the nearest hundredth.

| Pentagon                                                                                                                                                                                                         | Hexagon                                                                                                                                                                                                        |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  <p><math>m = 11.76</math> cm</p> <p><math>d = 32.36</math> cm</p> <p><math>a = 13.76</math> cm</p> <p>Side lengths = 20 cm</p> |  <p><math>h = 16.02</math> cm</p> <p><math>d = 18\frac{1}{2}</math> cm</p> <p>Side lengths = <math>9\frac{1}{4}</math> cm</p> |
| <b>Area of Pentagon</b><br>$688$ cm <sup>2</sup>                                                                                                                                                                 | <b>Area of Hexagon</b><br>$222.28$ cm <sup>2</sup>                                                                                                                                                             |



Students with visual impairments or those struggling with decomposing shapes may be distracted by the many markings on the shapes in the table. Consider drawing a larger shape with dimensions for question 3 and using string or tape to label the dimensions.



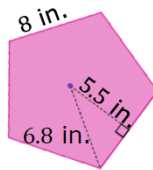
## 7.7.4 Your Turn!

**Component Overview:** Students practice independently decomposing regular polygons to find their areas. Students also analyze two samples of students' work. The students used different methods to decompose the regular hexagon. However, both methods result in the same area.



### 7.7.4 Your Turn!

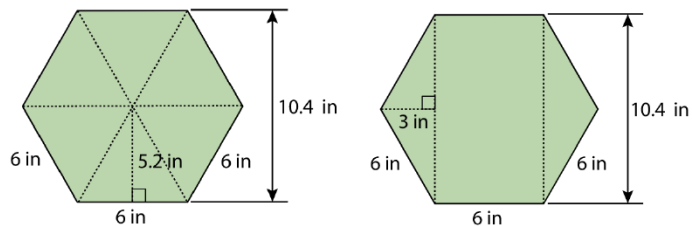
1. A regular pentagon is shown with lengths in inches (in.).



What is the area of the pentagon?

$110 \text{ in}^2$

2. Lin and Andre used different methods to find the area of a regular hexagon with 6 in. sides. Lin decomposed the hexagon into six identical, equilateral triangles. Andre decomposed the hexagon into a rectangle and two triangles.



- a. Explain why Lin and Andre will find the same area of the hexagon.

*Answers will vary. Lin and Andre will find the same area because decomposing the hexagon into different shapes does not change the area of the hexagon.*



*Students may misidentify the height of the triangle.*



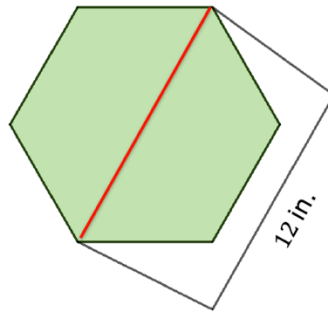
*Consider having students outline the triangles and quadrilateral in different colors to distinguish between the shapes.*

## 7.7.4 Your Turn! (continued)

- b. Find the area of the hexagon using both Lin's and Andre's methods.

| Lin's Method        | Andre's Method      |
|---------------------|---------------------|
| $93.6 \text{ in}^2$ | $93.6 \text{ in}^2$ |

- c. Decompose the shape differently than Lin and Andre did.



- d. Using the measurements given in all three figures, find the area by finding the sum of your decomposed parts.

$$93.6 \text{ in}^2$$



Prompt students to make a conjecture about the relationship between the number of sides in the regular polygon and the number of congruent triangles into which it can be decomposed. Ask students how this relationship can be used to create a general formula for the area of any regular polygon. Remind students to test their conjectures on a variety of different regular polygons.

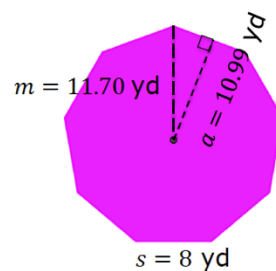
## 7.7.5 Wrap-Up: Error Analysis

**Component Overview:** Students identify and correct an error in a sample of a student's work. Consider using data from this formative assessment to determine which students need remediation before the next lesson.



### 7.7.5 Wrap-Up: Error Analysis

- Vernon found the area of the nonagon. His work is shown.



Students with visual impairments may struggle to read the dimensions on the nonagon or Vernon's work. Consider typing Vernon's work and enlarging the diagram of the nonagon.

#### Vernon's Work

|                             |                          |
|-----------------------------|--------------------------|
| Area of Triangle            | Area of Nonagon          |
| $A = \frac{1}{2}(8)(11.70)$ | $A = 18(46.8)$           |
| $A = 46.8 \text{ yd}^2$     | $A = 842.4 \text{ yd}^2$ |

- Explain to Vernon what error(s) he made in his work.

Answers will vary. Vernon made the errors of using the wrong height and multiplying the area by 18.

- What is the area of the nonagon?

395.64  $\text{yd}^2$